

Comparing two periods of high inflation: the Great Inflation and the 2020s inflation

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Abstract

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JEL:

1 Introduction

Inflation targeting has made inflation one of the most closely monitored macroeconomic indicators. When inflation rises sharply, attention shifts from headline inflation alone to the breadth, persistence and propagation of price increases. In 2021 at Jackson Hole, Powell (2021) played down the rise in inflation down as narrow and likely temporary, describing it as “largely the product of a relatively narrow group of goods and services that have been directly affected by the pandemic.” By 2022 that assessment had changed: Powell (2022) observed that “high inflation has continued to spread through the economy,” and, drawing on the lessons of the 1970s, warned that the anticipation of high inflation can become “entrenched in the economic decisionmaking of households and businesses.” This suggests for monetary policy two questions are central as inflation rises. First, are price increases concentrated in a small number of categories, or have they become broad-based across the household expenditure basket? A relative-price shock affecting a limited set of goods can often be looked through; a generalised increase across many categories is harder to dismiss Bryan and Cecchetti, 1994. Second, has inflation become self-reinforcing through wages and inflation expectations? If higher prices raise wage demands and inflation expectations, and those wages and expectations then feed back into prices, inflation may become more persistent and costly to reduce.

This paper asks whether the dependence structure among prices, wages and inflation expectations changes when inflation is high. In particular, we ask whether high-inflation episodes are

characterised by stronger contemporaneous co-movement across prices, stronger lagged propagation from one price category to another, or stronger feedback between prices, wages and expectations. We study these questions by comparing two major inflation episodes: the Great Inflation of the 1970s and the COVID-era inflation surge of 2021–22. Much of the conventional understanding of high-inflation dynamics is informed by the 1970s. The pandemic-era surge provides a modern, post-inflation-targeting episode in which inflation became sufficiently high to revisit those mechanisms.

The two episodes differed in important respects. The Great Inflation occurred in an economy more dependent on oil, with less anchored expectations and a different monetary-policy regime. The COVID-era surge, by contrast, reflected a combination of pandemic-related supply constraints, large shifts in goods demand, shipping disruption and subsequent energy and services pressures. These differences suggest that the two episodes may have broadened through different channels. We therefore organise the analysis around four hypotheses. The first two concern the broadening of inflation across prices and its interaction with wages:

H1 (CPI broadening). As inflation rises, connectedness among disaggregated CPI categories increases, particularly in the upper tail of the inflation distribution.

H2 (Wage–price interaction). The connectedness between wages and prices is stronger during high-inflation states than during normal-inflation states.

The next hypothesis distinguishes the mechanism through which inflation broadened in the two episodes:

H3 (Episode mechanism). The COVID-era surge was characterised primarily by contemporaneous connectedness, consistent with a large common shock, whereas the Great Inflation was characterised more by lagged propagation across prices.

Finally, we test whether institutional changes, especially the adoption of inflation targeting and the associated anchoring of expectations, altered the role of expectations in the inflation process:

H4 (Expectations channel). Inflation expectations were a stronger source of inflation propagation during the Great Inflation than during the COVID-era surge.

Finally, we consider whether the breadth of price changes shapes the conduct of monetary policy. There is an argument that monetary policy should respond to broad-based, core inflation whilst ‘looking through’ idiosyncratic movements in individual components (relative price changes).¹ All else equal, then, when price changes are highly connected across the subcomponents of the CPI, and as such is less likely to reflect idiosyncratic price changes, such inflation may be harder to look through, and monetary policy makers may respond to it more strongly:

H5 (Connectedness and monetary policy). When the subcomponents of the CPI are more highly connected, monetary policy responds more strongly to inflation.

Further, a related channel operates through the interaction between inflation and inflation expectations. When realised and expected inflation become more tightly connected, the risk of a

¹See Goodfriend and King, 1997, p.276

self-reinforcing spiral rises, and policy makers may respond more aggressively to forestall it. In particular, when inflation is spilling over into inflation expectations, it suggests expectations have become more adaptive than anchored.

H6 (Connectedness and monetary policy II). When inflation spills over more strongly into inflation expectations, monetary policy responds more strongly to inflation.

Methodologically, the paper draws on the connectedness literature. Connectedness measures are most commonly used to quantify directional spillovers in financial systems (Diebold & Yilmaz, 2012), but they are well suited to the study of inflation because they provide a direct measure of how movements in one series are associated with movements in others. The closest prior contribution is Stenfors et al. (2026), who study connectedness among disaggregated CPI series for the United States, the United Kingdom and Japan since the late 1980s using the time-varying-parameter VAR framework of Antonakakis et al. (2020). We make three contributions relative to that work.

The first is *span*. We extend the sample back to the Great Inflation, allowing the inflation surge of the 1970s and the COVID-era surge to be compared within a common empirical framework. The second is *scope*. In addition to the six disaggregated CPI categories, we include wages and inflation expectations, allowing us to examine wage–price interactions and the expectations channel directly. The third is *methodological*. We replace the TVP-VAR with the pseudo- R^2 quantile connectedness approach of Shahzad et al. (2026). This is important for two reasons. First, the method measures connectedness across the inflation distribution, rather than only at the conditional mean or median, allowing the dependence structure in high-inflation states to differ from that in normal times.² Second, the method decomposes connectedness into contemporaneous and lagged components. This distinction is central to our empirical design: it lets us separate broad contemporaneous co-movement, consistent with a common shock, from sequential propagation across prices, wages and expectations—precisely the distinction underlying H3.

Our central finding is that the dependence structure among prices does change with the level of inflation: cross-category connectedness rises roughly threefold from the median to the high-inflation tail, and the rise is increasingly accounted for by the *lagged* component rather than contemporaneous co-movement, with shelter and core services the central lagged nodes. The two episodes broadened differently — the COVID surge chiefly through contemporaneous co-movement, consistent with a common shock, and the Great Inflation more through lagged dependence. A prominent narrative that is not borne out is that inflation expectations lead: in high-inflation episodes they read as adaptive, following realised inflation. We stress throughout that connectedness measures statistical association and lead–lag dependence, not identified causal transmission; our claims concern how that dependence structure shifts across inflation states, rather than what causes what. The remainder of the paper proceeds as follows. Section 2 describes the data; Section 3 sets out the method; and Section 4 reports the tests of the hypotheses.

²A rolling-window TVP-VAR could in principle track the two episodes through time, but it still measures connectedness at the centre of the conditional distribution within each window; it cannot isolate behaviour in the tail.

2 Data

Our headline analysis studies the interconnectedness among the six monthly CPI subcomponent series drawn from the FRED database that together form a complete, non-overlapping decomposition of headline CPI. Each is expressed as a month-on-month percentage change and the sample covers February 1967 to May 2026. The six subcomponents are:

- Core goods (excluding food)
- Core services (excluding shelter)
- Shelter
- Food
- Household energy
- Gasoline

These six follow the standard expenditure-class decomposition: energy is separated into gasoline and household energy services, and core services into shelter and core services excluding shelter. The shelter series is available from FRED, but the core-services-ex-shelter series we construct as:

$$\% \Delta \text{CoreServ}^{\text{ex-shelter}} = \frac{\% \Delta \text{CoreServ} - w \% \Delta \text{Shelter}}{1 - w}, \quad (1)$$

where $w = 0.60$ is shelter's share of core services.³

We separate shelter from other core services because shelter inflation is driven by distinct dynamics — long lease durations and the imputed owners' -equivalent-rent methodology — and may behave differently from other services prices.

Our subsequent analyses use three additional series, again from FRED. For the overall Consumer Price Index, we use the Consumer Price Index for All Urban Consumers, All Items, transformed into a month-on-month percentage change to obtain headline inflation.⁴ Wage growth is measured using the average hourly earnings of production and nonsupervisory employees in the total private sector (AHETPI, dollars per hour, seasonally adjusted), expressed as a month-on-month percentage change. We use this series rather than the broader all employees measure because it is the longest-running monthly earnings series, beginning in 1964, and therefore spans the Great Inflation. A long span of data is needed for the for comparative analysis of inflation between the Great Inflation and the COVID Inflation. The broader all-employees series begins only in 2006.

³The results are insensitive to this choice: connectedness and the directional structure are stable for $w \in [0.50, 0.65]$ (reported in the robustness analysis).

⁴We omit the overall CPI from the subcomponent system because it is constructed from those very subcomponents, so any connectedness found would in part reflect a mechanical accounting relationship rather than economic co-movement.

Finally, we use the University of Michigan median expected change in prices over the next twelve months (MICH, percent), available monthly from January 1978. The monthly series therefore misses the 1973–77 period, the early un-anchoring phase of the Great Inflation, which is not ideal for a hypothesis about expectations during that episode. The Michigan measure is, however, available back to 1960 at a quarterly frequency, so we also run the expectations analysis quarterly. Our Great-Inflation expectations claims rest on that longer quarterly series (MICH_QTR); the monthly analysis uses MICH from 1978.

3 Methodology

3.1 Pseudo-quantile R^2 connectedness

3.1.1 Setup

Let $\mathbf{y}_t = (y_{1,t}, \dots, y_{K,t})'$ collect the K time series we study. In one application this system is the month-on-month percentage change of the six CPI subcomponents; in another it is wage growth, inflation expectations and headline inflation.

For each series $y_{k,t}$ in turn, we ask how much of its variation can be attributed to (i) variation of the remaining series in the system at time t , $\mathbf{y}_{-k,t}$, and (ii) the first p lags of all the time series in the whole system, $(\mathbf{y}'_{t-1}, \dots, \mathbf{y}'_{t-p})'$. The first source captures *contemporaneous* dependence: categories that move together within the month (or the quarter), while the second captures *lagged* dependence, that is, delayed propagation from one series to another over time. The distinction between the two is what allows us to separate a common shock from sequential pass-through.

A natural starting point is the goodness of fit of a regression of $y_{k,t}$ on this regressor set. The difficulty is that the regressors are themselves mutually correlated, so the regression R^2 cannot be partitioned unambiguously into the portion owed to each one.

Genizi (1993) proposes an order-invariant decomposition of the regression R^2 that distributes it across correlated regressors through the symmetric (eigenvalue) square root of their correlation matrix. Let $\mathbf{R}_{xx}$ denote the correlation matrix of the regressors, with eigendecomposition $\mathbf{R}_{xx} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}'$, where $\mathbf{V}$ holds the eigenvectors and $\mathbf{\Lambda}$ the eigenvalues. Defining the symmetric square root $\mathbf{C}$, the vector of decomposed R^2 shares is

$$\mathbf{R}_{xx} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}' = \mathbf{C}\mathbf{C}', \quad (2)$$

$$\mathbf{C} = \mathbf{V}\mathbf{\Lambda}^{1/2}\mathbf{V}', \quad (3)$$

$$\mathbf{R}^{2,d} = [\mathbf{C}^2 (\mathbf{C}^{-1}\mathbf{r}_{yx})^2]', \quad (4)$$

where $\mathbf{r}_{yx}$ is the vector of correlations between the dependent variable and each regressor. Each entry of $\mathbf{R}^{2,d}$ is the share of the regression R^2 owed uniquely to a single regressor, net of the correlation it shares with the others; the entries are non-negative and sum to the regression R^2 .

Using the above decomposition with Pearson correlations as inputs, Balli et al. (2023) show how contemporaneous and lagged connectedness can be quantified at the centre of the distribution. Because, however, our interest is in how dependence behaves when inflation is high, we use the adaptation of Shahzad et al. (2026), which replaces the Pearson correlations with the quantile correlation of Choi and Shin (2022); this allows contemporaneous and lagged connectedness to be measured at any chosen quantile τ .

The specifics of the Shahzad et al. (2026) measure are as follows. The quantile correlation of Choi and Shin (2022) between two variables, $r_{1,2,\tau}$, is substituted for R_{xx} and r_{yx} in equations 2 and 4. For each time series in turn, equation 4 then returns a vector $\mathbf{R}_{\tau}^{2,d}$ that allocates the regression's overall pseudo- R^2 across its regressors at that τ .⁵ The first $K - 1$ entries record the contemporaneous contributions of the other series; the remaining entries record the lagged contributions of all series in the system.

For a system of K variables, applying (4) to each of the K equations in that system and stacking the results yields the connectedness matrix

$$\mathbf{R}_{\tau}^{2,d} = \begin{bmatrix} \mathbf{R}_{\tau,1}^{2,d} \\ \vdots \\ \mathbf{R}_{\tau,K}^{2,d} \end{bmatrix} = [\mathbf{R}_{lag=0,\tau}^{2,d} \mid \mathbf{R}_{lag=1,\tau}^{2,d} \mid \cdots \mid \mathbf{R}_{lag=p,\tau}^{2,d}], \quad (5)$$

in which row k indexes the dependent (or the receiving) variable.⁶

The total connectedness index at quantile τ , TCI_{τ} , is the average across the K series of the off-diagonal $R^{2,d}$ shares, and splits into a contemporaneous and a lagged component,

$$\text{TCI}_{\tau} = \text{TCI}_{\tau}^C + \text{TCI}_{\tau}^L, \quad (6)$$

with

$$\text{TCI}_{\tau}^C = \frac{1}{K} \sum_{k=1}^K \sum_{j \neq k} \mathbf{R}_{lag=0,\tau,k,j}^{2,d}, \quad \text{TCI}_{\tau}^L = \frac{1}{K} \sum_{k=1}^K \sum_{j=1}^K \sum_{l=1}^p \mathbf{R}_{lag=l,\tau,k,j}^{2,d}, \quad (7)$$

where $\mathbf{R}_{lag=l,\tau,k,j}^{2,d}$ is the share of series k 's variation contributed by series j at lag l . Throughout, connectedness measures are reported as percentages (shares of the R^2 multiplied by 100). Equivalently, TCI_{τ} is the average of the directional FROM — or, identically, TO — measures defined below.

Directional measures in a similar spirit to those put forward by Diebold and Yilmaz (2012) can also be calculated. The total connectedness *received* by series k (its FROM measure) sums the shares

⁵Quantile regressions do *not* have an R^2 in the conventional (least-squares) sense, hence the term “pseudo”.

⁶So that the equations align, the k th position of $\mathbf{R}_{\tau,k}^{2,d}$ is set to zero, since $y_{k,t}$ is not regressed on its own contemporaneous value.

contributed by all other series, at every lag, into row k ;

$$\text{FROM}_{k,\tau} = \sum_{j \neq k} R_{lag=0,\tau,k,j}^{2,d} + \sum_{j=1}^K \sum_{l=1}^p R_{lag=l,\tau,k,j}^{2,d}. \quad (8)$$

The total connectedness *transmitted* by series j to other series (its TO measure) sums the shares it contributes, at every lag, down column j :

$$\text{TO}_{j,\tau} = \sum_{k \neq j} R_{lag=0,\tau,k,j}^{2,d} + \sum_{k=1}^K \sum_{l=1}^p R_{lag=l,\tau,k,j}^{2,d}. \quad (9)$$

The net directional connectedness of series j ,

$$\text{NET}_{j,\tau} = \text{TO}_{j,\tau} - \text{FROM}_{j,\tau}, \quad (10)$$

is positive when j is, on balance, a transmitter of connectedness and negative when it is a net receiver; averaging FROM, equivalently TO, across series returns TCI_τ .

We prefer this measure to other approaches to quantile connectedness, such as that of Ando et al. (2022). Those measures propagate the non-zero-mean residuals from quantile regressions through a generalised forecast-error variance decomposition that treats them as zero-mean shocks. As Shahzad et al. (2026) show, this inflates measured connectedness in both tails.

4 Results

All connectedness measures reported below are percentages (shares of the R^2 , multiplied by 100).

4.1 H1: inflation broadens in the high tail

Our first hypothesis is **H1 (CPI broadening)**: as inflation rises, connectedness among disaggregated CPI categories increases, particularly in the upper tail of the inflation distribution. Table 1 reports the overall connectedness at different levels of τ over our whole sample period (the and its decomposition into contemporaneous and lagged connectedness; Table 1 reports the same by quantile, monthly (Panel A) and quarterly (Panel B). Overall connectedness rises about threefold from the median to the high tail (from 11.8 to 35.2 at monthly frequency); both the contemporaneous and lagged components increase, but slightly more of the rise comes from the lagged component (23.3 vs 11.9 at $\tau = 0.9$). This indicates that high-inflation states are characterised not only by stronger common movements across prices, but also by more pronounced sequential propagation across categories. This result also holds if quarterly data are used (see Panel B).

Table ?? in the appendix show the contemporaneous dependency between CPI subcomponents at $\tau = 0.9$. The rows indicate FROM connectedness so Food gets 0.2 units from Gas and 1.6 from

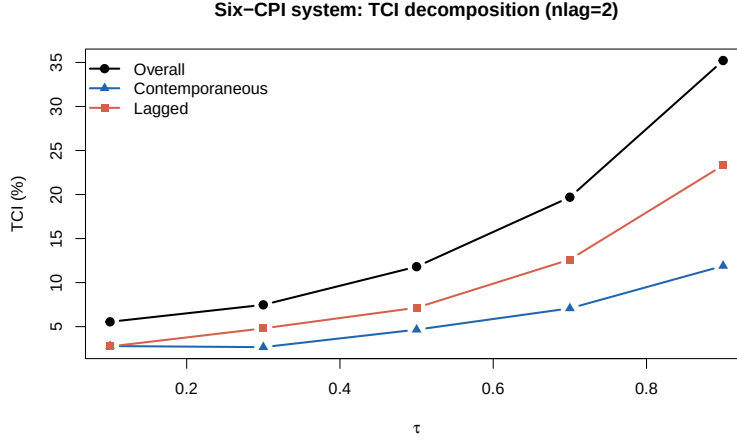


Fig. 1 Overall connectedness (black) and its decomposition into contemporaneous (blue) and lagged (red) components, by inflation quantile τ (monthly, $n_{\text{lag}} = 2$).

Energy; the columns indicate TO connectedness, so Food gives 0.4 to gasoline and 1.6 household energy. The strongest contemporaneous connectedness is recorded between household energy and gasoline (in both directions), and core goods and core services with shelter, again in both directions. The bidirectional nature of these suggests they represent common shocks. Higher nominal incomes and low interest rates could increase demand for houses, good and services for example. Table [] shows the same for lagged connectedness. In line with the results in Table 1 we see, generally, connectedness is stronger with a lag. WE see that shelter, services and gasoline are a net transmitter of non-contemporaneous connectedness, where as Food, core goods and household energy are net receivers [need to revise this].

4.2 H3: how the two episodes broadened

Our most distinctive result concerns *how* inflation broadened in each episode. Table 2 reports how high-tail ($\tau = 0.9$) connectedness changes when each episode is added to the calm 1983–2019 core — estimating the model with the Great Inflation included (but COVID dropped) and with COVID included (but the Great Inflation dropped). Both contemporaneous and lagged connectedness rise by more in the Great Inflation, consistent with its higher inflation level. More tellingly, the Great-Inflation increase comes mainly through the *lagged* component, whereas the COVID increase is almost entirely *contemporaneous* — its lagged marginal is in fact negative. The two components have a natural reading: contemporaneous connectedness is categories moving together within the period, as a common shock hitting all prices at once would produce; lagged connectedness is one category’s past movement being associated with another’s current movement, the signature of a sequential, cross-category process. The COVID pattern is thus consistent with a common shock (pan-

Table 1 H1 — Inflation broadens into the high tail

<i>Panel A. Monthly total connectedness by quantile</i>			
τ	Overall	Contemporaneous	Lagged
0.1	5.5	2.8	2.8
0.3	7.5	2.7	4.8
0.5	11.8	4.7	7.2
0.7	19.7	7.1	12.6
0.9	35.2	11.9	23.3

<i>Panel B. Quarterly total connectedness by quantile</i>			
τ	Overall	Contemporaneous	Lagged
0.1	16.9	8.3	8.6
0.3	17.6	6.8	10.8
0.5	27.7	10.7	17.1
0.7	55.7	21.6	34.1
0.9	86.8	33.8	52.9

Notes: Pseudo-quantile R^2 (Genizi/nearPD) connectedness (%) among the six CPI subcomponents; Overall = Contemporaneous + Lagged, $n_{\text{lag}} = 2$. Panel A is monthly 1967–2026 (broadening ratio $\text{TCI}(0.9)/\text{TCI}(0.5) = 2.98\times$); Panel B quarterly 1967Q2–2026Q2 (ratio $3.13\times$). Both frequencies show the tail rise, increasingly driven by the lagged component. Point estimates; bootstrap confidence intervals for the six-CPI system are to be added.

demic supply constraints and shipping disruption), and the Great-Inflation pattern with a slower, sequential build-up — supporting H3. This distinction is precisely the object the contemporaneous/lagged decomposition is built to measure, and it links to the wage channel examined next. The contrast is sharp at monthly frequency (Panel A); at quarterly frequency (Panel B) the calm-era base connectedness is already high and lagged-dominated, so the episode marginals are small and less sharply signed, and we read the monthly result as the primary evidence.

4.3 Self-reinforcement: wages and expectations

The next two hypotheses ask whether inflation became self-reinforcing through wages and expectations. Table 3 shows that the connectedness between wages and the overall CPI rises with the quantile (H2), and that at higher quantiles the lagged inflation–wage association strengthens. The pattern holds at quarterly frequency (Panel C) and is concentrated in the Great-Inflation-inclusive subperiod (Panel B): wage–price connectedness at $\tau = 0.9$ is 25.2 in 1964–2019 against only 3.6 post-1983. Wage–price dependence is thus, like broadening itself, a high-inflation-regime phenomenon.

For expectations (H4), Table 4 reports the net directional connectedness of inflation expectations. Pooled over the full sample, expectations are net transmitters in the upper tail — more connected *to* inflation than they receive *from* it — but this reflects the anchored 1983–2019 period, where expectations lead (NET = +29.1 at $\tau = 0.9$). Conditional on a high-inflation episode the sign reverses: in

Table 2 H3 — COVID broadened contemporaneously; the Great Inflation through lagged propagation

<i>Panel A. Monthly: marginal $\Delta TCI(0.9)$ vs. the 1983–2019 core</i>				
Episode	Δ Overall	Δ Contemp.	Δ Lagged	Lagged %
GreatInflation (1967-82)	+14.3	+5.0	+9.3	64.9
COVID (2020-26)	+1.1	+1.9	-0.8	-70.6

<i>Panel B. Quarterly: marginal $\Delta TCI(0.9)$ vs. the 1983–2019 core</i>				
Episode	Δ Overall	Δ Contemp.	Δ Lagged	Lagged %
GreatInflation (1967-82)	+3.2	+3.8	-0.6	-17.8
COVID (2020-26)	-18.6	-4.6	-14.0	75.4

Notes: Each episode’s marginal contribution to high-tail ($\tau = 0.9$) connectedness relative to the calm 1983–2019 core (nested subtraction), split into contemporaneous and lagged; six-CPI system, Panel A monthly, Panel B quarterly. At monthly frequency the Great Inflation broadened mainly through the *lagged* (propagation) channel while COVID broadened contemporaneously (its lagged marginal is negative). At quarterly frequency the calm-era base connectedness is already high and lagged-dominated, so the episode marginals are small and less sharply signed.

Table 3 H2 — Wage–price connectedness is stronger when inflation is high

<i>Panel A. Two-variable wage–CPI system (monthly, full sample), by quantile</i>			
	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.9$
Wage–CPI total connectedness (TCI)	10.8	16.6	21.6
lagged Wages→CPI	5.0	7.7	11.5
lagged CPI→Wages	6.2	9.2	16.1

<i>Panel B. By subperiod (monthly)</i>			
	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.9$
<i>1964–2019 (with Great Inflation)</i>			
Wage–CPI total connectedness (TCI)	13.3	18.4	25.2
lagged Wages→CPI	6.0	9.1	15.5
lagged CPI→Wages	5.9	9.0	14.9
<i>1983–2026 (post-Great-Inflation)</i>			
Wage–CPI total connectedness (TCI)	1.2	1.8	3.6
lagged Wages→CPI	0.2	0.2	1.3
lagged CPI→Wages	0.9	1.7	3.0

<i>Panel C. Quarterly (full sample), by quantile</i>			
	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.9$
Wage–CPI total connectedness (TCI)	25.7	32.6	53.4
lagged Wages→CPI	14.6	19.3	35.1
lagged CPI→Wages	13.0	18.5	23.8

Notes: Two-variable system of wage growth (AHETPI) and overall CPI (CPIAUCSL); pseudo-quantile R^2 connectedness (%), Genizi estimator, $n_{\text{lag}} = 2$. Wage–price connectedness rises sharply into the high-inflation tail at all frequencies, and is far stronger in the Great-Inflation-inclusive subperiod (1964–2019) than post-1983. In the high tail the lagged components dominate. Quarterly ($n_{\text{lag}} = 2$) spans a six-month horizon.

the high tail of both the Great Inflation (NET = −20.2) and COVID (−6.7) expectations follow realised inflation, consistent with adaptive expectations. The Great-Inflation figure uses the quarterly Michigan series (from 1960), which covers the 1973–77 un-anchoring the monthly series misses. Table 5 decomposes the link by direction: in the full sample the lagged expectations→inflation term is the larger, but in the Great Inflation the lagged inflation→expectations term dominates at higher τ (reaching 50.7 at $\tau = 0.9$, against 27.5 in the reverse direction) — the adaptive channel. COVID shows the same pattern, less pronounced.

Table 4 H4 — Do inflation expectations lead or follow? Net directional connectedness

Sample	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.9$
Full (1960–2026)	+4.8	+13.7	+10.5
Great Inflation (1967–82)	+0.5	-2.2	-20.2
Core (1983–2019)	+3.0	+8.8	+29.1
COVID (2020–26)	-5.5	-4.1	-6.7

Notes: Net directional connectedness of quarterly Michigan inflation expectations (MICH_QTR, from 1960) versus overall CPI in a two-variable system, $n_{\text{lag}} = 2$. NET = TO − FROM; > 0 means expectations lead (transmit), < 0 means they follow. Pooled, expectations lead in the upper tail — but this reflects the anchored 1983–2019 core (NET +29.1 at $\tau = 0.9$); conditional on a high-inflation episode the sign reverses (Great Inflation −20.2, COVID −6.7), consistent with adaptive expectations. The COVID window (25 quarters) is short.

Table 5 H4 — Contemporaneous and lagged connectedness between inflation and expectations

Sample	τ	Inflation → Expectations		Expectations → Inflation	
		Contemp.	Lagged	Contemp.	Lagged
Full (1960–2026)	0.5	24.2	23.7	27.4	25.4
	0.7	20.9	28.5	27.7	35.4
	0.9	19.5	33.6	26.2	37.3
Great Inflation (1967–82)	0.5	28.4	27.7	27.6	29.0
	0.7	22.4	28.9	23.0	26.2
	0.9	22.5	50.7	25.5	27.5
Core (1983–2019)	0.5	17.9	8.8	22.8	6.8
	0.7	14.9	6.2	21.6	8.2
	0.9	21.3	11.9	27.5	34.7
COVID (2020–26)	0.5	0.3	6.1	0.4	0.5
	0.7	1.6	11.9	1.7	7.7
	0.9	0.8	7.8	1.3	0.6

Notes: Pairwise pseudo-quantile R^2 connectedness (%) in a two-variable quarterly system of Michigan inflation expectations (MICH_QTR, from 1960) and overall CPI (CPIAUCSL), $n_{\text{lag}} = 2$ (a six-month lag horizon). “Inflation → Expectations” is the share of *expectations*’ variation explained by CPI, split into contemporaneous and lagged; “Expectations → Inflation” is the reverse. The *lagged* Inflation → Expectations term (the adaptive channel) dominates in the high tail of both episodes — most strongly in the Great Inflation (50.7 at $\tau = 0.9$) — whereas in the calm 1983–2019 period the lagged Expectations → Inflation term is the larger. Dependence, not identified causation.

To test H5 and H6 we proceed in three steps. First, for each connectedness system we compute the pseudo-quantile R^2 connectedness measure over rolling windows, using both 60- and 120-month windows and evaluating it at $\tau = 0.5, 0.7$ and 0.9 so as to characterise connectedness at the centre as well as in the upper tail of the distribution. This yields, for each window length and quantile, a monthly series of total connectedness, C_t .

Second, we classify each month into a high- or low-connectedness regime. Rather than imposing an ad hoc threshold, we let the data determine the regimes by fitting the two-state Markov-switching model of Hamilton (1989) to C_t . Specifically, we treat connectedness as a first-order autoregression whose intercept, autoregressive coefficient and error variance all switch between two latent states, $s_t \in \{\text{low}, \text{high}\}$, that evolve as a first-order Markov chain. The AR(1) specification captures the well-known persistence of the connectedness series whilst allowing both its level and its volatility to differ across regimes. We estimate the model by maximum likelihood, recover the smoothed probability that each month belongs to each state, and assign every month to its most probable state; the state with the higher mean connectedness is labelled the ‘high’ regime, so that $\text{High}_t = 1$ for months in that state and zero otherwise.

Third, we estimate an inertial Taylor rule augmented with a regime interaction:

$$i_t = \alpha + \rho i_{t-1} + \beta_\pi (\pi_t - \pi^*) + \delta (\text{High}_t \cdot (\pi_t - \pi^*)) + \beta_x x_t + \varepsilon_t, \quad (11)$$

where i_t is the Wu–Xia (Wu & Xia, 2016) shadow policy rate, π_t is twelve-month CPI inflation, $\pi^* = 2$ per cent is the assumed inflation target, and x_t is the unemployment gap.⁷ The coefficient ρ captures interest-rate smoothing Clarida et al., 2000, p.151-152; one motivation for this specification is that under uncertainty central banks only partially adjust to the interest rate warranted based on current economic conditions. In equation 11 β_π the baseline response to the inflation gap, and β_x the response to unemployment. The object of interest is δ : it measures the *additional* response to inflation in the high-connectedness regime. If H5 and H6 are true, then $\delta > 0$: monetary policy responds more strongly to inflation when price changes (H5) or inflation and inflation expectations (H6) are more highly connected. All inference is based on Newey–West HAC standard errors.

Table ?? reports estimates of equation (11) in which the high- and low-connectedness regimes are identified from the connectedness of the six CPI subcomponents, measured over a 120-month rolling window. We re-estimate the rule over several sample start dates because the monetary-policy reaction function is known to have shifted over time, with different Federal Reserve chairs placing different weight on inflation (Clarida et al., 2000); estimating over later subsamples guards against attributing such governor-preference changes in the rule to the connectedness regime.

Two readings of this evidence are possible. On the first, the state dependence is real; but recent: it emerges only in the samples beginning in 1993 and 2003, consistent with modern central banks,

⁷he unemployment gap is the difference between the civilian unemployment rate and the Congressional Budget Office’s estimate of the natural (non-cyclical) rate of unemployment, both obtained from FRED (series UNRATE and NROU). Because the CBO series is published quarterly, we linearly interpolate it to monthly frequency.

operating under explicit or implicit inflation targets. being more attentive to the breadth of price changes than their predecessors. That the effect appears only at $\tau = 0.9$ reinforces this reading, suggesting that policymakers respond more strongly to inflation only once connectedness is concentrated in the upper tail, amongst those components already experiencing large price increases. This interpretation has some plausibility. A second, more sceptical reading is that the two significant coefficients are false positives. The interaction δ is estimated over twelve quantile-sample combinations, so at conventional levels one or two spurious rejections are to be expected by chance alone; that one of the two is significant only at the 10 per cent level lends this reading further weight. We therefore treat the result as no more than suggestive, and note below that it does not survive a shorter, 60-month estimation window.

Table shows connectedness between inflation expectations and inflation under two different measures of inflation. One the annual percentage change in inflation (panel A), one is the monthly change in inflation (panel B); we study the annual percentage change so it is one the same period as inflation expectations which are measured over a year. On that measure only $\tau = 0.7$ connectedness shows any change in the monetary policy reaction function for inflation; this does not survive if monthly CPI growth is used.

Table 6 reports the Taylor rule estimates when the connectedness regime is identified from the inflation-expectations system, under two measures of inflation: the twelve-month (annual) percentage change in the CPI (Panel A) and the month-on-month change (Panel B). We prefer the annual measure, as it is defined over the same twelve-month horizon as the Michigan expectations series. On that measure the interaction δ is positive and statistically significant only at $\tau = 0.7$, where it is significant across every sample start date, with δ ranging from 0.04 to 0.06 indicating a stronger inflation response in high-connectedness months. This result does not survive when month-on-month CPI inflation is used instead: in Panel B δ is statistically insignificant throughout. Again not overly strong support for the hypothesis that connectedness is taken into account when setting monetary policy.

A limitation of the preceding analysis is that, at a monthly frequency, connectedness is estimated with only two lags, so lagged spillovers are captured over a two-month horizon. This restriction is necessary: within the short rolling windows the underlying vector autoregression must remain estimable, and the implied variance-decomposition matrix must be well-conditioned and positive semi-definite. Additional lags rapidly exhaust the available degrees of freedom and degrade the connectedness estimates. The concern, however, is that the price spirals and broad subcomponent co-movements that might concern policymakers could build over a horizon longer than two months. We therefore re-estimate connectedness at a quarterly frequency, again with two lags, which extends the horizon over which spillovers can propagate to roughly six months. The Taylor rule continues to be estimated at a monthly frequency; a given month is assigned to the high- or low-connectedness regime according to the state of the quarter in which it falls.

Table 8 reports the Taylor-rule estimates for the six CPI subcomponents when the connectedness

Table 6 Subcomponent connectedness

	Earliest (1977)	1983	1993	2003
$\tau = 0.5$				
Intercept	0.16** (2.1)	0.10* (1.7)	0.04 (0.9)	0.04 (1.1)
$\rho (i_{t-1})$	0.95*** (54.3)	0.96*** (62.3)	0.99*** (96.5)	0.98*** (95.4)
$\beta_\pi (\pi - \pi^*)$	0.06** (2.3)	0.02 (1.0)	-0.03 (-0.6)	0.02 (1.1)
$\Delta\beta_\pi^{\text{High}}$	0.01 (0.3)	0.03 (1.4)	0.07 (1.3)	0.03 (1.4)
$\beta_x (\text{ugap})$	-0.04** (-2.4)	-0.03* (-1.8)	-0.02 (-1.5)	-0.03** (-2.3)
n	590	518	398	278
high-conn. share	0.40	0.33	0.63	0.15
long-run β_π (low/high)	1.19/1.31	0.53/1.41	-2.91/3.92	0.94/2.52
$\tau = 0.7$				
Intercept	0.16** (2.2)	0.11* (1.8)	0.04 (0.9)	0.04 (1.0)
$\rho (i_{t-1})$	0.95*** (54.7)	0.97*** (61.6)	0.99*** (98.4)	0.98*** (97.5)
$\beta_\pi (\pi - \pi^*)$	0.03 (1.1)	-0.01 (-0.4)	-0.02 (-0.5)	0.02 (1.3)
$\Delta\beta_\pi^{\text{High}}$	0.04 (1.5)	0.06 (1.6)	0.06 (1.2)	0.03 (1.3)
$\beta_x (\text{ugap})$	-0.04** (-2.5)	-0.03* (-1.8)	-0.02 (-1.5)	-0.03** (-2.2)
n	590	518	398	278
high-conn. share	0.62	0.62	0.58	0.16
long-run β_π (low/high)	0.59/1.31	-0.41/1.29	-2.20/3.67	1.34/2.81
$\tau = 0.9$				
Intercept	0.17** (2.2)	0.10 (1.6)	0.03 (0.9)	0.04 (1.3)
$\rho (i_{t-1})$	0.95*** (55.4)	0.97*** (61.9)	0.99*** (116.0)	0.98*** (104.4)
$\beta_\pi (\pi - \pi^*)$	0.04 (1.2)	0.03 (1.2)	0.01 (0.4)	0.01 (0.7)
$\Delta\beta_\pi^{\text{High}}$	0.03 (1.1)	0.02 (0.6)	0.04** (2.0)	0.04* (1.8)
$\beta_x (\text{ugap})$	-0.04** (-2.3)	-0.03* (-1.7)	-0.02* (-1.8)	-0.03** (-2.6)
n	590	518	398	278
high-conn. share	0.45	0.37	0.26	0.31
long-run β_π (low/high)	0.67/1.33	0.82/1.32	0.45/3.52	0.57/2.42

Notes: Estimates of equation (11). Columns are sample start dates; each panel a quantile τ . Newey–West HAC t -statistics in parentheses; *, **, *** denote significance at 10, 5 and 1 per cent. $\Delta\beta_\pi^{\text{High}}$ is the interaction δ ; the final row reports the implied long-run response $\beta_\pi / (1 - \rho)$ in the low and high regimes.

Table 7 Expectations connectedness and the policy reaction function: 120-month window

Panel A. Matched: MICH and 12-month CPI inflation			
	Earliest (1987)	1993	2003
$\tau = 0.5$			
Intercept	0.05 (1.4)	0.03 (0.9)	0.04 (1.3)
$\rho (i_{t-1})$	0.98*** (112.3)	0.98*** (111.3)	0.98*** (99.2)
$\beta_\pi (\pi - \pi^*)$	0.02 (1.1)	0.02 (1.1)	0.03 (1.6)
$\Delta\beta_\pi^{\text{High}}$	0.01 (0.7)	0.03 (1.5)	0.01 (0.5)
$\beta_x (\text{ugap})$	-0.03*** (-2.6)	-0.02** (-2.1)	-0.03** (-2.5)
n	459	398	278
high-conn. share	0.23	0.21	0.12
long-run β_π (low/high)	1.01/1.73	1.42/3.08	1.48/2.05
$\tau = 0.7$			
Intercept	0.06* (1.8)	0.04 (1.2)	0.05 (1.4)
$\rho (i_{t-1})$	0.97*** (125.9)	0.98*** (111.2)	0.97*** (94.9)
$\beta_\pi (\pi - \pi^*)$	0.01 (0.8)	0.02 (1.3)	0.02 (1.2)
$\Delta\beta_\pi^{\text{High}}$	0.05** (2.6)	0.04** (2.3)	0.06*** (3.5)
$\beta_x (\text{ugap})$	-0.03*** (-3.0)	-0.02** (-2.2)	-0.03*** (-2.7)
n	459	398	278
high-conn. share	0.41	0.41	0.41
long-run β_π (low/high)	0.46/2.41	1.05/3.36	0.64/2.85
$\tau = 0.9$			
Intercept	0.06* (1.8)	0.04 (1.1)	0.05 (1.4)
$\rho (i_{t-1})$	0.98*** (121.8)	0.98*** (110.1)	0.98*** (99.8)
$\beta_\pi (\pi - \pi^*)$	0.01 (0.6)	0.02 (0.7)	0.05*** (3.3)
$\Delta\beta_\pi^{\text{High}}$	0.03 (1.2)	0.03 (1.3)	-0.03 (-1.0)
$\beta_x (\text{ugap})$	-0.04*** (-3.0)	-0.02** (-2.0)	-0.03** (-2.5)
n	459	398	278
high-conn. share	0.59	0.61	0.42
long-run β_π (low/high)	0.48/1.66	0.82/2.70	2.08/0.92
Panel B. Mismatched: MICH and month-on-month CPI inflation			
	Earliest (1987)	1993	2003
$\tau = 0.5$			
Intercept	0.05 (1.4)	0.02 (0.6)	0.03 (0.8)
$\rho (i_{t-1})$	0.98*** (113.9)	0.99*** (110.0)	0.99*** (99.1)
$\beta_\pi (\pi - \pi^*)$	0.00 (0.1)	-0.00 (-0.1)	0.04*** (3.3)
$\Delta\beta_\pi^{\text{High}}$	0.03 (1.1)	0.05* (1.8)	-0.05* (-1.8)
$\beta_x (\text{ugap})$	-0.03** (-2.5)	-0.02 (-1.6)	-0.03** (-2.0)
n	458	397	277
high-conn. share	0.46	0.42	0.55
long-run β_π (low/high)	0.13/1.59	-0.29/4.54	3.34/ - 0.35
$\tau = 0.7$			
Intercept	0.05 (1.4)	0.04 (0.9)	0.04 (1.1)
$\rho (i_{t-1})$	0.98*** (113.7)	0.98*** (109.1)	0.98*** (91.9)
$\beta_\pi (\pi - \pi^*)$	0.04 (1.6)	0.03 (1.1)	0.03 (1.4)
$\Delta\beta_\pi^{\text{High}}$	-0.01 (-0.5)	0.01 (0.2)	0.00 (0.1)
$\beta_x (\text{ugap})$	-0.03** (-2.4)	-0.02* (-1.9)	-0.03** (-2.2)
n	458	397	277
high-conn. share	0.47	0.39	0.30
long-run β_π (low/high)	1.71/1.02	1.74/2.22	1.60/1.74
$\tau = 0.9$			
Intercept	0.05 (1.4)	0.04 (0.9)	0.05 (1.2)
$\rho (i_{t-1})$	0.98*** (117.2)	0.98*** (105.9)	0.98*** (93.3)
$\beta_\pi (\pi - \pi^*)$	0.03 (1.6)	0.03 (1.3)	0.04* (1.7)
$\Delta\beta_\pi^{\text{High}}$	-0.00 (-0.2)	0.02 (1.0)	-0.00 (-0.2)
$\beta_x (\text{ugap})$	-0.03*** (-2.6)	-0.02* (-2.0)	-0.03** (-2.2)
n	458	397	277
high-conn. share	0.40	0.35	0.36
long-run β_π (low/high)	1.34/1.15	1.57/2.57	1.73/1.58

Notes: Estimates of equation (11) in which the connectedness regime is identified from the inflation-expectations system over a 120-month rolling window. Panel A pairs the Michigan one-year inflation expectation with 12-month CPI inflation; Panel B pairs it with month-on-month CPI inflation. Columns are sample start dates; within each panel the three blocks are the quantiles $\tau = 0.5, 0.7$ and 0.9 . Newey–West HAC t -statistics in parentheses; *, **, *** denote significance at 10, 5 and 1 per cent. $\Delta\beta_\pi^{\text{High}}$ is the interaction δ ; the final row of each block reports the implied long-run response $\beta_\pi / (1 - \rho)$ in the low and high regimes.

regime is based on connectedness measured at quarterly frequency. The evidence is notably stronger than under the monthly measure. Of the eight estimates of the interaction δ at $\tau = 0.5$ and $\tau = 0.7$ —four sample start dates at each quantile—six are positive and statistically significant at the 10 per cent level or better (most at 5 per cent or better), indicating a stronger inflation response in high-connectedness months. At $\tau = 0.9$, by contrast, only one of the four estimates is statistically significant.

Table 8 Subcomponent connectedness: Quarterly

	Earliest (1977)	1983	1993	2003
$\tau = 0.5$				
Intercept	0.17** (2.2)	0.12* (1.9)	0.03 (0.9)	0.04 (1.2)
$\rho(i_{t-1})$	0.95*** (52.4)	0.96*** (61.7)	0.99*** (132.4)	0.98*** (112.7)
$\beta_\pi(\pi - \pi^*)$	0.05** (2.3)	0.01 (0.4)	0.01 (0.4)	0.01 (0.6)
$\Delta\beta_\pi^{\text{High}}$	0.03 (1.5)	0.06*** (2.7)	0.06*** (3.2)	0.05*** (3.0)
$\beta_x(\text{ugap})$	-0.05** (-2.6)	-0.04** (-2.4)	-0.02** (-2.3)	-0.03*** (-3.0)
n (months)	589	517	397	277
high-conn. share	0.32	0.28	0.13	0.19
long-run β_π (low/high)	0.91/1.42	0.14/1.55	0.38/4.41	0.39/2.85
$\tau = 0.7$				
Intercept	0.16** (2.1)	0.11* (1.8)	0.02 (0.5)	0.03 (0.8)
$\rho(i_{t-1})$	0.95*** (54.2)	0.96*** (61.1)	0.99*** (127.8)	0.98*** (109.5)
$\beta_\pi(\pi - \pi^*)$	0.02 (1.3)	0.01 (0.7)	0.00 (0.1)	0.01 (0.5)
$\Delta\beta_\pi^{\text{High}}$	0.05** (2.2)	0.03 (1.4)	0.05** (2.2)	0.04* (1.8)
$\beta_x(\text{ugap})$	-0.05*** (-2.6)	-0.04** (-2.4)	-0.02** (-2.0)	-0.02** (-2.1)
n (months)	589	517	397	277
high-conn. share	0.42	0.34	0.20	0.31
long-run β_π (low/high)	0.46/1.35	0.37/1.24	0.14/5.14	0.60/3.06
$\tau = 0.9$				
Intercept	0.16** (2.4)	0.11* (1.7)	0.03 (0.7)	0.03 (1.0)
$\rho(i_{t-1})$	0.95*** (57.9)	0.96*** (59.3)	0.99*** (111.0)	0.98*** (105.0)
$\beta_\pi(\pi - \pi^*)$	0.07*** (3.5)	0.02 (0.7)	0.01 (0.6)	0.01 (0.5)
$\Delta\beta_\pi^{\text{High}}$	-0.03 (-1.4)	0.03 (1.3)	0.03 (1.3)	0.04* (1.7)
$\beta_x(\text{ugap})$	-0.04** (-2.3)	-0.04** (-2.0)	-0.02* (-1.7)	-0.02** (-2.3)
n (months)	589	517	397	277
high-conn. share	0.40	0.44	0.39	0.36
long-run β_π (low/high)	1.41/0.89	0.40/1.25	0.83/3.13	0.60/2.86

Notes: Estimates of equation (11). Columns are sample start dates; each panel a quantile τ . Newey–West HAC t -statistics in parentheses; *, **, *** denote significance at 10, 5 and 1 per cent. $\Delta\beta_\pi^{\text{High}}$ is the interaction δ ; the final row reports the implied long-run response $\beta_\pi/(1 - \rho)$ in the low and high regimes.

Table ?? repeats the analysis for the inflation–expectations system under quarterly connectedness; an added advantage of this is quarterly inflation expectations are available back to 1960. Using year-on-year inflation, matched to the one-year horizon of the Michigan survey, any evidence of a stronger inflation response is confined to $\tau = 0.5$, where the interaction δ is positive and statistically significant in the 1993 and 2003 samples. At $\tau = 0.7$ and $\tau = 0.9$ it is statistically insignificant, and in one specification significantly negative. Aggregating connectedness to a quarterly frequency therefore does not strengthen the evidence that inflation–expectations con-

nectedness shifts the monetary-policy reaction function.⁸

Table 9 Expectations connectedness: Quarterly

	Extended [†] (1969)	Earliest (1987)	1993	2003
$\tau = 0.5$				
Intercept	0.20*** (2.9)	0.05 (1.3)	0.03 (0.8)	0.04 (1.1)
$\rho(i_{t-1})$	0.94*** (53.8)	0.98*** (109.4)	0.99*** (118.7)	0.98*** (112.6)
$\beta_\pi(\pi - \pi^*)$	0.08*** (3.2)	0.02 (1.3)	0.01 (0.6)	0.01 (0.3)
$\Delta\beta_\pi^{\text{High}}$	-0.05** (-2.3)	0.01 (0.6)	0.05*** (3.0)	0.04* (1.9)
$\beta_x(\text{ugap})$	-0.06*** (-2.8)	-0.03** (-2.5)	-0.02** (-2.1)	-0.03*** (-2.8)
n (months)	674	460	397	277
high-conn. share	0.16	0.27	0.31	0.42
long-run β_π (low/high)	1.26/0.40	1.04/1.66	0.61/3.88	0.30/2.61
$\tau = 0.7$				
Intercept	0.16*** (2.8)	0.06 (1.5)	0.05 (1.3)	0.05 (1.3)
$\rho(i_{t-1})$	0.95*** (67.1)	0.98*** (104.8)	0.98*** (105.2)	0.97*** (90.7)
$\beta_\pi(\pi - \pi^*)$	0.03 (1.0)	0.03 (1.6)	0.04*** (2.7)	0.01 (0.3)
$\Delta\beta_\pi^{\text{High}}$	0.07** (2.5)	-0.01 (-0.2)	-0.05* (-1.8)	0.03 (1.6)
$\beta_x(\text{ugap})$	-0.05*** (-2.7)	-0.03** (-2.6)	-0.03** (-2.1)	-0.03** (-2.2)
n (months)	674	460	397	277
high-conn. share	0.56	0.22	0.21	0.90
long-run β_π (low/high)	0.55/1.79	1.28/0.95	2.11/ - 0.47	0.24/1.59
$\tau = 0.9$				
Intercept	0.18*** (2.9)	0.05 (1.5)	0.04 (0.9)	0.04 (1.2)
$\rho(i_{t-1})$	0.95*** (64.8)	0.98*** (109.3)	0.98*** (101.3)	0.98*** (97.0)
$\beta_\pi(\pi - \pi^*)$	0.05 (1.3)	0.03 (1.4)	0.02 (0.9)	0.01 (0.6)
$\Delta\beta_\pi^{\text{High}}$	0.01 (0.2)	-0.01 (-0.2)	0.01 (0.4)	0.03 (1.0)
$\beta_x(\text{ugap})$	-0.05*** (-2.9)	-0.03** (-2.5)	-0.02* (-1.9)	-0.03** (-2.6)
n (months)	674	460	397	277
high-conn. share	0.64	0.30	0.27	0.24
long-run β_π (low/high)	0.92/1.08	1.39/1.13	1.53/2.25	0.84/2.39

Notes: Estimates of equation (11). Columns are sample start dates; each panel a quantile τ . Newey–West HAC t -statistics in parentheses; *, **, *** denote significance at 10, 5 and 1 per cent.

$\Delta\beta_\pi^{\text{High}}$ is the interaction δ ; the final row reports the implied long-run response $\beta_\pi/(1 - \rho)$ in the low and high regimes.

[†] The Extended column uses the quarterly Michigan survey (available from 1960); the remaining columns use the monthly survey (from 1978) aggregated to quarters.

A Connectedness matrices

Tables 10–15 report the full pairwise connectedness structure among the six CPI subcomponents at $\tau = 0.9$ ($n_{\text{lag}} = 2$), separately for the contemporaneous and lagged blocks, for the full sample and the two subperiods (1967–1982 and 1983–2026). In each matrix, cell (i, j) is the share of row i 's variation attributed to column j ; FROM is the row sum (received), TO the column sum (transmitted), and NET = TO–FROM. The contemporaneous block is close to symmetric (small NET), whereas the lagged block carries the directional structure — shelter and core services are the net lagged transmitters. Gasoline is weakly connected to the other categories: its variation is dominated by

⁸This conclusion holds if inflation is measured as quarterly percentage change as well

energy-market (crude-oil) movements that lie outside the CPI system, so it sits at the periphery of the network.

Table 10 Contemporaneous connectedness among CPI subcomponents (full sample, 1967–2026, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	—	0.2	1.6	4.6	5.4	1.2	12.9
Gasoline	0.4	—	4.9	0.1	0.1	0.1	5.6
HH energy	1.6	4.4	—	1.8	1.0	1.2	9.9
Core goods	4.6	0.0	1.8	—	6.4	2.5	15.3
Shelter	4.9	0.0	0.5	6.3	—	5.8	17.5
Core svc. ex-sh.	0.8	0.1	1.2	2.3	5.7	—	10.1
TO	12.2	4.7	10.0	15.2	18.5	10.8	
NET	-0.7	-0.9	+0.0	-0.2	+1.0	+0.7	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\text{lag}} = 2$. Cell (i, j) is the share of row i 's variation attributed to column j . FROM = row sum (received); TO = column sum (transmitted); NET = TO – FROM (> 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO.

Table 11 Lagged connectedness among CPI subcomponents (full sample, 1967–2026, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	23.5	0.3	5.5	6.6	13.1	4.6	30.1
Gasoline	1.4	11.3	1.4	0.3	0.3	0.3	3.6
HH energy	2.8	2.8	10.2	2.6	1.9	6.2	16.3
Core goods	6.8	0.4	3.1	29.3	5.6	9.9	25.8
Shelter	12.0	0.1	1.3	6.5	26.6	12.8	32.7
Core svc. ex-sh.	3.5	0.9	3.6	7.1	16.2	15.4	31.4
TO	26.5	4.5	14.9	23.1	37.1	33.8	
NET	-3.6	+0.9	-1.4	-2.8	+4.4	+2.4	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\text{lag}} = 2$. Cell (i, j) is the share of row i 's variation attributed to column j . FROM = row sum (received); TO = column sum (transmitted); NET = TO – FROM (> 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO (and shown for the lagged block as the own-lag/persistence term).

B Goods-to-services sequencing: a lag-window illustration

A frequently cited feature of the 2021–22 surge is that goods inflation led services inflation. Our framework recovers this, but only once the measurement window is long enough — which is itself instructive about connectedness measurement. Table 16 reports, on the post-1983 sample at $\tau = 0.9$, the net lagged connectedness from core goods to core services. Under the monthly two-lag dependence (Genizi) metric the sign is slightly negative (-1.2): a two-month window is too short to span a multi-month goods-then-services sequence. The LASSO predictive metric, which admits longer lags, recovers the ordering ($+1.3$ at two lags, $+5.9$ at six, where core goods becomes a net

Table 12 Contemporaneous connectedness among CPI subcomponents (1967–1982, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	—	4.9	1.9	8.5	2.9	1.3	19.6
Gasoline	3.6	—	6.9	4.0	4.7	7.9	27.0
HH energy	1.5	18.9	—	3.1	1.6	2.8	28.1
Core goods	4.3	11.5	2.3	—	4.6	4.8	27.5
Shelter	2.2	16.2	2.0	4.8	—	10.5	35.8
Core svc. ex-sh.	2.9	16.6	1.9	4.1	5.4	—	31.0
TO	14.6	68.2	15.0	24.6	19.3	27.3	
NET	-5.0	+41.2	-13.1	-2.9	-16.5	-3.7	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\text{lag}} = 2$. Cell (i, j) is the share of row i 's variation attributed to column j . FROM = row sum (received); TO = column sum (transmitted); NET = TO – FROM (> 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO.

Table 13 Lagged connectedness among CPI subcomponents (1967–1982, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	31.5	2.1	23.2	9.4	3.8	6.0	44.4
Gasoline	22.3	24.4	10.9	4.3	8.4	2.8	48.6
HH energy	15.8	8.1	15.9	19.7	4.8	7.3	55.8
Core goods	6.9	4.8	17.4	34.7	5.0	3.4	37.6
Shelter	15.5	6.3	2.8	20.4	16.5	2.5	47.5
Core svc. ex-sh.	16.2	9.5	3.6	18.6	5.3	15.5	53.3
TO	76.7	30.9	57.9	72.4	27.4	21.9	
NET	+32.3	-17.7	+2.1	+34.9	-20.1	-31.4	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\text{lag}} = 2$. Cell (i, j) is the share of row i 's variation attributed to column j . FROM = row sum (received); TO = column sum (transmitted); NET = TO – FROM (> 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO (and shown for the lagged block as the own-lag/persistence term).

Table 14 Contemporaneous connectedness among CPI subcomponents (1983–2026, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	—	0.2	6.3	0.3	2.2	1.0	9.9
Gasoline	0.4	—	4.0	0.1	3.5	0.3	8.2
HH energy	6.5	3.8	—	1.1	0.2	0.4	12.0
Core goods	0.3	0.1	1.0	—	0.4	4.1	5.8
Shelter	2.2	2.9	0.2	0.4	—	2.8	8.5
Core svc. ex-sh.	1.0	0.2	0.4	4.2	2.8	—	8.6
TO	10.4	7.2	11.8	6.0	9.1	8.4	
NET	+0.5	-1.0	-0.2	+0.2	+0.6	-0.1	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\text{lag}} = 2$. Cell (i, j) is the share of row i 's variation attributed to column j . FROM = row sum (received); TO = column sum (transmitted); NET = TO – FROM (> 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO.

transmitter). Table 17 confirms it at quarterly frequency, where a two-lag block spans six months: the lagged goods→services term then exceeds the reverse at $\tau = 0.5$ (9.9 vs 4.4) and $\tau = 0.7$ (11.2

Table 15 Lagged connectedness among CPI subcomponents (1983–2026, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	14.6	1.6	9.2	2.6	4.8	0.7	18.9
Gasoline	3.6	10.3	3.0	0.0	0.8	0.6	8.0
HH energy	4.6	3.8	10.8	0.8	0.8	1.2	11.2
Core goods	2.6	1.6	1.4	30.7	0.7	3.0	9.3
Shelter	7.5	0.5	0.5	0.5	19.9	8.3	17.4
Core svc. ex-sh.	3.1	8.1	3.1	2.1	8.8	5.7	25.1
TO	21.4	15.6	17.3	6.0	15.8	13.9	
NET	+2.4	+7.6	+6.1	-3.3	-1.6	-11.3	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\text{lag}} = 2$. Cell (i, j) is the share of row i 's variation attributed to column j . FROM = row sum (received); TO = column sum (transmitted); NET = TO–FROM (> 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO (and shown for the lagged block as the own-lag/persistence term).

vs 7.1). The point is methodological — lag-window length materially affects what connectedness detects, which motivates the quarterly analysis used elsewhere — rather than a new fact about the episode.

Table 16 Goods-to-services sequencing under alternative metrics and lag lengths

Metric	Goods→services (lagged, net)	Core goods NET
Genizi dependence ($n_{\text{lag}}=2$)	-1.2	-3.4
LASSO predictive ($n_{\text{lag}}=2$)	+1.3	+3.4
LASSO predictive ($n_{\text{lag}}=6$)	+5.9	+12.9

Notes: “Goods→services (lagged, net)” is lagged connectedness from core goods to core services (shelter + core services ex-shelter) minus the reverse, at $\tau = 0.9$ on the ex-Great-Inflation sample (1983–2026); > 0 means goods lead. The two-lag Genizi dependence metric cannot see a multi-month sequence and tilts negative; the LASSO predictive metric confirms the sequencing, increasingly at longer lags. Predictive ordering, not identified causation.

References

- Ando, T., Greenwood-Nimmo, M., & Shin, Y. (2022). Quantile connectedness: Modeling tail behavior in the topology of financial networks. *Management Science*, 68(4), 2401–2431.
- Antonakakis, N., Chatziantoniou, I., & Gabauer, D. (2020). Refined measures of dynamic connectedness based on time-varying parameter vector autoregressions. *Journal of Risk and Financial Management*, 13(4), 84.
- Balli, F., Balli, H. O., Dang, T. H. N., & Gabauer, D. (2023). Contemporaneous and lagged R2 decomposed connectedness approach: New evidence from the energy futures market. *Finance Research Letters*, 57, 104168.

Table 17 Goods-to-services: contemporaneous and lagged connectedness (quarterly)

τ	Goods $\rightarrow$ Services		Services $\rightarrow$ Goods	
	Contemp.	Lagged	Contemp.	Lagged
0.5	3.3	9.9	3.4	4.4
0.7	8.5	11.2	8.7	7.1
0.9	13.1	18.4	15.2	18.3

Notes: Pairwise pseudo-quantile R^2 connectedness (%) between core goods and core services (shelter + core services ex-shelter) on the ex-Great-Inflation quarterly sample (1983Q1–2026Q2, 174 quarters), $n_{\text{lag}} = 2$ (a six-month lag horizon). “Goods $\rightarrow$ Services” is the share of services’ variation explained by goods, split into contemporaneous and lagged; “Services $\rightarrow$ Goods” is the reverse. The *lagged* Goods $\rightarrow$ Services term is the “goods lead” propagation channel; it exceeds the lagged Services $\rightarrow$ Goods term at every quantile (by 4.0 at $\tau = 0.7$), consistent with goods leading services. Six months is the horizon the monthly two-lag metric could not span. Dependence, not identified causation.

- Bryan, M. F., & Cecchetti, S. G. (1994). Measuring core inflation [In N.G. Mankiw (ed.), *Monetary Policy*, University of Chicago Press, 195–215]. *Monetary Policy* (NBER).
- Choi, J.-E., & Shin, D. W. (2022). Quantile correlation coefficient: A new tail dependence measure. *Statistical Papers*, 63(4), 1075–1104.
- Clarida, R., Galí, J., & Gertler, M. (2000). Monetary policy rules and macroeconomic stability: Evidence and some theory. *Quarterly Journal of Economics*, 115(1), 147–180.
- Diebold, F. X., & Yilmaz, K. (2012). Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1), 57–66.
- Genizi, A. (1993). Decomposition of R^2 in multiple regression with correlated regressors. *Statistica Sinica*, 3(2), 407–420.
- Goodfriend, M., & King, R. G. (1997). The new neoclassical synthesis and the role of monetary policy. In B. S. Bernanke & J. J. Rotemberg (Eds.), *Nber macroeconomics annual 1997* (pp. 231–296, Vol. 12). MIT Press.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.
- Powell, J. H. (2021). Monetary policy in the time of COVID-19 [Speech at “Macroeconomic Policy in an Uneven Economy,” economic policy symposium sponsored by the Federal Reserve Bank of Kansas City, Jackson Hole, Wyoming, August 27].
- Powell, J. H. (2022). Monetary policy and price stability [Speech at “Reassessing Constraints on the Economy and Policy,” economic policy symposium sponsored by the Federal Reserve Bank of Kansas City, Jackson Hole, Wyoming, August 26].
- Shahzad, S. J. H., Ryan, M., & Gabauer, D. (2026). Pseudo-quantile R^2 connectedness [companion methodology paper] [Working paper].

- Stenfors, A., Shabani, M., Gabauer, D., & Toporowski, J. (2026). Decomposing the rate of inflation: Forecast-based connectedness among cpi components. *Economic Modelling*, 163, 107708.
- Wu, J. C., & Xia, F. D. (2016). Measuring the macroeconomic impact of monetary policy at the zero lower bound. *Journal of Money, Credit and Banking*, 48(2-3), 253–291.